



Prognosis of Juvenile myoclonic epilepsy with eye-closure sensitivity

Carina Gonçalves Pedroso Uchida^a, Kelly Cristina de Carvalho^a,
Mirian Salvadori Bittar Guaranha^a, Laura Maria Figueiredo F. Guilhoto^a,
Gerardo Maria de Araújo Filho^b, Elza Márcia Targas Yacubian^{a,*}

^a Unidade de Pesquisa e Tratamento das Epilepsias, Department of Neurology and Neurosurgery, Universidade Federal de São Paulo (UNIFESP), São Paulo, SP, Brazil

^b Department of Psychiatry, Universidade Federal de São Paulo (UNIFESP), São Paulo, SP, Brazil

ARTICLE INFO

Keywords:

Juvenile myoclonic epilepsy
Eye closure sensitivity
Eyelid myoclonia
Prognosis

ABSTRACT

Purpose: Eye closure sensitivity (ECS) has been described as a reflex trait in juvenile myoclonic epilepsy (JME). However, there is no consensus regarding its significance on prognosis. The aim of this study is to clarify the long-term impact of ECS documented by a clinical interview and a video-EEG neuropsychological protocol (VNPP) in a series of 133 JME patients.

Methods: Data from 22 JME patients with ECS confirmed by a VNPP (Group 1) were compared with those of 20 JME patients without any reflex traits (Group 2). They were followed for a mean of 8.21 years (SD = ± 5.044). The frequency of seizures was assessed using a diary. Except for photosensitivity (PS), any other reflex traits occurrence, drugs/alcohol abuse intake, noncompliance, and Jeavons syndrome, were considered exclusion criteria.

Results: Group 1 had a lower age at epilepsy onset ($p = 0.028$), higher incidence of febrile seizures (13.6%), and familial history of epilepsy ($p = 0.023$). Only 18.2% had self-perception of eyelid myoclonia (EM) (kappa coefficient = 0.193), which persisted in 77.3% of patients. Limb myoclonia, tonic-clonic seizures (TCS) and/or myoclonic-tonic-clonic seizures (MTCS), as well as absences were more frequent ($p = 0.015$; $p = 0.013$; $p = 0.011$, respectively) in Group 1. PS did not influence frequency of EM ($p = 1.0$), absences ($p = 0.648$), or TCS/MTCS ($p = 0.934$). Psychiatric comorbidities were not different between groups.

Conclusions: ECS is related to a worse outcome regarding control of all seizure types, persistence of EM, and higher frequency of limb myoclonia, as well as the total number of TCS and/or MTCS.

1. Introduction

Juvenile myoclonic epilepsy (JME) is a heterogeneous genetic generalized epilepsy (GGE) syndrome starting around puberty, comprising 5–10% of all epilepsies in large cohorts [1]. Myoclonic jerks of upper extremities, predominantly occurring after awakening, without loss of consciousness, constitute its main feature. Other seizure types are absences and tonic-clonic seizures (TCS) [2,3] and/or myoclonic-tonic-clonic seizures (MTCS) according to the 2017 ILAE classification of seizures [4,5].

Eye closure sensitivity (ECS) is a unique EEG phenomenon described as the non-incidental appearance of transient epileptic abnormalities following the closure of the eyes for 1–4 s during a standard EEG recording [6], eyelid myoclonia (EM) being its clinical expression.

EM, with or without absences, triggered by eye closure, were firstly reported by Radovici et al. [7]. The description of Jeavons [8] allowed

sufficient criteria for syndromic diagnosis, so that the eponym of Jeavons syndrome (JS), a condition of childhood onset, EM and absences (EMA), associated with brief bilateral spike and wave activity, spontaneous absences, possible occurrence of generalized TCS, and tendency to persist over the lifespan, has been considered a separate entity among GGEs [9–12]. However, currently, it is still considered a seizure type by ILAE [13], which occurs in several epileptic conditions [14,15], including JME [16–24].

Reflex epileptic traits, such as praxis induction and visual stimuli triggering myoclonia, are set in Class II diagnostic criteria of JME [25]. These reflex traits, isolated or in clusters, might characterize different JME endophenotypes predisposing to varied therapeutic responses and prognoses [26].

Since there is no consensus regarding the significance ECS on JME prognosis (Fig. 1) [16,19–24], the aim of this study is to clarify the long-term impact of this reflex trait documented by a clinical interview

* Corresponding author.

E-mail address: yacubian@terra.com.br (E.M.T. Yacubian).

<https://doi.org/10.1016/j.seizure.2018.09.006>

Received 10 July 2018; Received in revised form 10 September 2018; Accepted 12 September 2018

1059-1311/ © 2018 British Epilepsy Association. Published by Elsevier Ltd. All rights reserved.

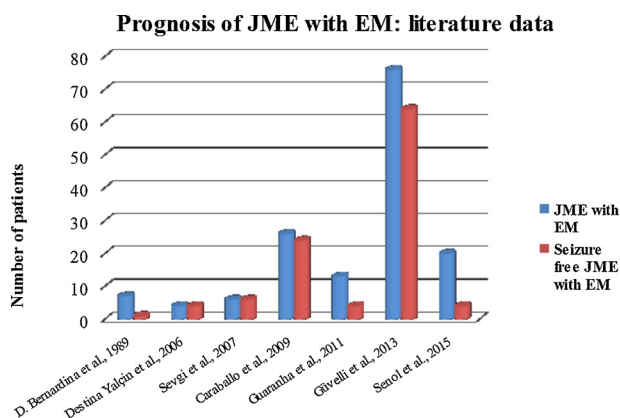


Fig. 1. Prognosis regarding seizure control in juvenile myoclonic epilepsy patients with eyelid myoclonia. Literature data. Legend: EM: eyelid myoclonia; JME: juvenile myoclonic epilepsy.

and a video-EEG neuropsychological protocol (VNPP) in a series of JME patients.

2. Methods

2.1. Clinical evaluation

To determine the endophenotype of reflex traits in 133 JME patients, we used a semi-structured interview based on a questionnaire. All patients had the unequivocal diagnosis of JME, based on electroclinical characteristics, including normal physical and neurological examinations, routine blood tests, brain imaging (CT/MRI), and generalized 4–6 Hz spike or polyspike-wave complexes, sometimes asymmetric, on a normal background in routine EEGs. They have been followed in the outpatient clinic of a tertiary center (Epilepsy Section, Department of Neurology and Neurosurgery, Universidade Federal de São Paulo, São Paulo, Brazil).

Initially, a clinical interview focused on myoclonic seizures occurrence, self-perception of EM, and their circadian distribution (exclusively or predominantly on awakening, and the period of occurrence during the day) was applied. It was followed by a questionnaire regarding precipitant factors of seizures, including three questions: (a) Have you noticed any situations or states which do cause you to have more seizures?; (b) Can you identify some precipitating factors on this list: stress, sleep deprivation, specific thoughts/concentration, flashing lights, performing hand activities and complex finger manipulation, playing games, calculation, speaking in public, alcohol intake, playing musical instruments, listening to music, writing, dancing, drawing, menses, and others?; (c) Can you identify some factors or situations that would stop or inhibit your seizures?

Psychiatric comorbidity was also analyzed through Schedule Clinical Interview for DSM-IV, Axis I (ICD-10), and/or “The Mini-International Neuropsychiatric Interview” (MINI) [27] and the State-Trait Anxiety Inventory (STAI), aiming to measure state (STAI-S) and trait (STAI-T) anxiety components [28]. The results of the STAI fitted between 38 and 42 were considered normal, whereas results lower than 38 denoted a depressive tendency, and results higher than 42 were considered indicative of susceptibility to anxiety.

2.2. Video-EEG neuropsychological protocol

Our VNPP was based on those reported by Matsuoka et al. [29] and Guaranha et al. [30]. After obtaining informed consent, all patients had a 4–6 h video-EEG monitoring, comprehending a research protocol approved by our Ethics Committee. Video-EEG was recorded using a digital equipment (Biologic 1, software Ceeograph 1), with the 10–20

International Electrode System, in addition to perioral and deltoid electrodes, besides palpebral electrodes (a piezoelectric polyvinylidene fluoride film transducer, length = 2.0 cm, width = 0.5 cm, and weight = 4.5 g) [31]. The protocol was postponed for those who had presented TCS and/or MTCS over the last 48 h; antiepileptic drugs (AEDs) were maintained in all.

After sleeping only for up to 4 h, the patients had 30 min of awake EEG recording, used as baseline, followed by at least 5-min of awake tracing with closed eyes, and 5-min of each one of the following nine methods of activation: reading silently and aloud in native (Portuguese) and in a non-fluent (English) language; speaking aloud (about their seizures, a very stressful task); two praxis tasks (Rubik’s cube, Jenga tower, Conundrum’s cube, or Pyramid puzzle); blink reflex (by threat); and hyperventilation (HV). Intermittent photic stimulation (IPS) with eyes closure, closed eyes, and opened eyes was evaluated according to Kasteleijn-Nolst Trenité et al. [32], with and without blue special Carl Zeiss filter lenses F-452 [33]. Opened/closed eyes were evaluated through command, self-paced, passive (the eyelids were opened and closed by the fingers of the examiner), and in the dark (using diving goggles sealed to light). Patients had at least 10 min of recording during a meal (breakfast or lunch). During the recordings, the focus of the camera was at the eyes of the subjects, allowing evaluation of EM occurrence.

We defined three types of EM: a) flicker, a slight rhythmic vibration of the eyelids without tonic spasm; b) flutter, large and rhythmic eyelid movements, like the flapping of a butterfly, during which the upper and lower eyelid move away from each other; and c) jerk, a sudden, rhythmic and quick tremor accompanied by a brief tonic spasm of the eyelids [34,35]. The EEG findings were grouped as “spiky posterior alpha”; posterior spike and wave complexes; and diffuse spike and wave complexes [35].

We considered activation as the increase by twice of the number of seizures or epileptic discharges (ED), in comparison to the baseline [29,30].

2.3. Inclusion and exclusion criteria

Group 1 included 22 JME patients with ECS and EM confirmed by VNPP. Only 4 out of these 22 patients had reported EM in the previous clinical interview and in the questionnaire of precipitant factors. Their data were compared to those of 20 JME patients without any reflex epileptic trait in the same conditions (Group 2). Any other reflex traits occurrence, except for photosensitivity (PS), drugs/alcohol abuse intake, noncompliance, and JS, were considered exclusion criteria.

2.4. Follow-up

All patients were followed-up for 2–17 years (mean = 8.21 yrs., SD = $\pm$ 5.044). Seizure types and precipitant factors, AEDs therapy, and treatment adherence were observed. All patients were oriented to avoid sleep deprivation and alcohol consumption. They received sodium valproate (VPA) as first choice drug, in mono or polytherapy, and other AEDs considered reasonably effective in JME treatment, such as topiramate, lamotrigine, phenobarbital, and benzodiazepines. AEDs doses were titrated according to the clinical response and the adverse effects. The frequency of seizures was monitored by standard seizure calendars. For the comparative evaluation of AEDs administered, the concept of AED load was applied [36]. It was calculated by summing the ratios between the prescribed dose and the World Health Organization defined doses of each medication [37].

Myoclonia and absence frequencies were quantified as seizure days per month at the first clinical interview and currently. Regarding TCS and/or MTCS, the frequency per month was assessed at the first evaluation. At present, the sum of all seizures occurring over the last year of follow-up was considered. In addition, the total lifetime number of TCS and/or MTCS was estimated at the last evaluation.

Table 1
Demographic data.

	JME with eye-closure sensitivity	JME without any reflex trait	p
N	22	20	
Gender (females)	18 (81.8%)	12 (60.0%)	*0.111
Age (average; $\pm$ SD)	28.1 $\pm$ 11.13 y	32.9 $\pm$ 10.68 y	**0.189
19–30 y	13 (59.1%)	10 (50.0%)	**0.412
31–40 y	7 (31.8%)	5 (25.0%)	
> 41 y	2 (9.1%)	5 (25.0%)	
Age at epilepsy onset (average; $\pm$ SD)	10.3 $\pm$ 4.46 y	13.3 $\pm$ 3.61 y	**0.028
Duration of epilepsy (average; $\pm$ SD)	17.7 $\pm$ 10.90 y	19.7 $\pm$ 11.27 y	**0.504
Follow-up (average; $\pm$ SD)	6.7 $\pm$ 4.49 y	9.3 $\pm$ 3.58 y	**0.029
Diagnosis delay (average; $\pm$ SD)	11.0 $\pm$ 8.79 y	10.4 $\pm$ 9.39 y	**0.666
Schooling (average; $\pm$ SD)	11.9 $\pm$ 3.42y	12.0 $\pm$ 3.70 y	**0.975
Estimated IQ (average; $\pm$ SD)	93.2 ($\pm$ 14.03)	97.7 ($\pm$ 11.96)	**0.561
STAI-T			
< 38	9 (40.9%)	7 (35.0%)	*0.963
Normal	4 (18.2%)	4 (20.0%)	
> 42	9 (40.9%)	9 (45.0%)	
STAI-S			
< 38	13 (59.1%)	7 (35.0%)	*0.194
Normal	6 (27.3%)	6 (30.0%)	
> 42	3 (13.6%)	7 (35.0%)	
Epilepsy in family	18 (81.8%)	9 (45.0%)	*0.023
first grade	5 (22.7%)	2 (10.0%)	*0.414
second grade	8 (36.4.0%)	2 (10.0%)	*0.071
third grade	7 (31.8%)	5 (25.0%)	*0.738
Febrile seizures	3 (13.6%)	0 (0.0%)	
Previous inappropriate use of AEDs	12 (54.5%)	11 (55.0%)	*1.0
Seizures			
at morning	22 (100.0%)	17 (85.0%)	*0.099
in the afternoon	2 (9.1%)	0 (0%)	*0.153
during the evening	4 (18.2%)	1 (5.0%)	
at night	0 (0%)	1 (5.0%)	

Legend: AEDs: antiepileptic drugs; JME: juvenile myoclonic epilepsy. *Fisher's exact test; **Mann-Whitney U test.

Bold values = $p < 0.05$.

2.5. Statistics

Comparison between Groups 1 and 2 was performed by the non-parametric Mann-Whitney U test for quantitative variables. The Fisher's exact test was applied for qualitative variables. The kappa coefficient was used to evaluate agreement between the questionnaire and VNPP findings. A p -value < 0.05 was considered statistically significant.

3. Results

Groups were homogeneous with respect to gender, age, duration of epilepsy, previous use of inappropriate AEDs, neuropsychological findings, and circadian distribution of seizures (Table 1).

The median age at epilepsy onset was 9 years old. Group 1 patients had a lower age at the epilepsy onset ($p = 0.028$), and a higher incidence of family history of seizures ($p = 0.023$). Three patients of this group had febrile seizures.

The epilepsy onset was different in the two groups. The patients of Group 1 had an earlier onset of epilepsy ($p = 0.028$). Regarding group 1 patients, absences and EM were more prevalent ($p = 0.016$) and more frequent (0.011) as the initial seizure type ($p = 0.006$).

Instead, the majority of the patients of Group 2 had epilepsy inaugurated by TCS and/or MTCS ($p < 0.001$) and limb myoclonia. Age at the epilepsy onset also influenced seizure patterns in Group 1, as the later the beginning, the more frequent the limb myoclonia ($p = 0.007$)

(Fig. 2).

In addition, among 13 patients with PS, 8 (61.6%) started having seizures before 9 years of age ($p = 0.031$).

3.1. Questionnaire

Among the general precipitants of seizures, sleep deprivation was reported by all patients of Group 1 and by 19 out 20 (95.0%) patients of Group 2 ($p = 0.476$). Stress was reported by 20 out 22 Group 1 patients (90.9%), and just two of whose had normal STAIT ($p = 0.026$).

With respect to specific triggers, 17 of the 22 (77.3%) patients of Group 1 reported environmental PS (agreement coefficient-kappa = 0.193), while only 4 (18.2%) had self-perception of EM at the interview (agreement coefficient-kappa = 0.021). Visuospatial tasks were referred to as a precipitant of seizures by 3 (13.6%) patients of Group 1, and verbal tasks (reading, speaking) by only 1 (4.5%).

It is worth mentioning that none of our patients had self-induced seizures.

3.2. Video-EEG findings

General VNPP data are shown in Table 2. While 15 of the 22 (68.2%) patients of Group 1 presented EM at baseline, no patient of Group 2 had limb myoclonia ($p < 0.001$). Interictal discharges on baseline were present in 20 patients (90.9%) of Group 1, and only in 3 (15.0%) of Group 2 ($p < 0.001$). Moreover, general precipitants such as sleep, HV, and awakening activated discharges and seizures in a higher proportion of JME patients with ECS ($p = 0.003$; $p = 0.025$; $p < 0.001$, respectively).

VNPP data regarding specific tasks inducing discharges and/or seizures in Group 1, as well as activation of discharges and/or seizures, are shown in Table 3 (Fig. 3).

3.3. Prognosis

Eighteen out 22 (81.8%) patients in Group 1, and 11 out 20 (55.0%) of Group 2 were not seizure free at the last visit ($p = 0.023$), and the frequency of limb myoclonia was higher in Group 1 ($p = 0.015$). EM persisted since the beginning of epilepsy, independently of the therapeutic scheme, in 17/22 patients (77.3%). At epilepsy onset, absences were both, more prevalent ($p = 0.016$) and more frequent (0.011) in Group 1.

TCS and/or MTCS prevailed in Group 1, at the beginning and currently, since 18/22 patients (81.8%) and 10/20 (50.0%), respectively, presented more than 10 convulsive seizures throughout life ($p = 0.013$).

In Group 1, PS did not influence the frequency of EM ($p = 1.0$), the absences ($p = 0.648$), or TCS and/or MTCS ($p = 0.934$).

Finally, there were no differences in ICD-10/MINI used to evaluate psychiatric comorbidity ($p = 1.0$; Table 4).

4. Discussion

Initially associated with JS [8–12], EM is considered a type of seizure by ILAE since 2001 [13], and can be expressed by JME patients [16–24]. Already documented by Janz in 1969 [38] and present in 5–20% of JME patients, electroencephalographic ECS has never been clinically characterized in detail in this epileptic syndrome [39], even though some reported that JS and JME may coexist in the same patient [19,34].

In this prospective study, we assessed the impact of the presence of ECS reflex trait in 22 JME patients versus 20 patients without any reflex trait, in the long-term (mean 8.21 yrs.) seizure control of all three seizure types.

In Group 1, patients had a lower age at the epilepsy onset and a higher incidence of a familial history of seizures. These data agree with

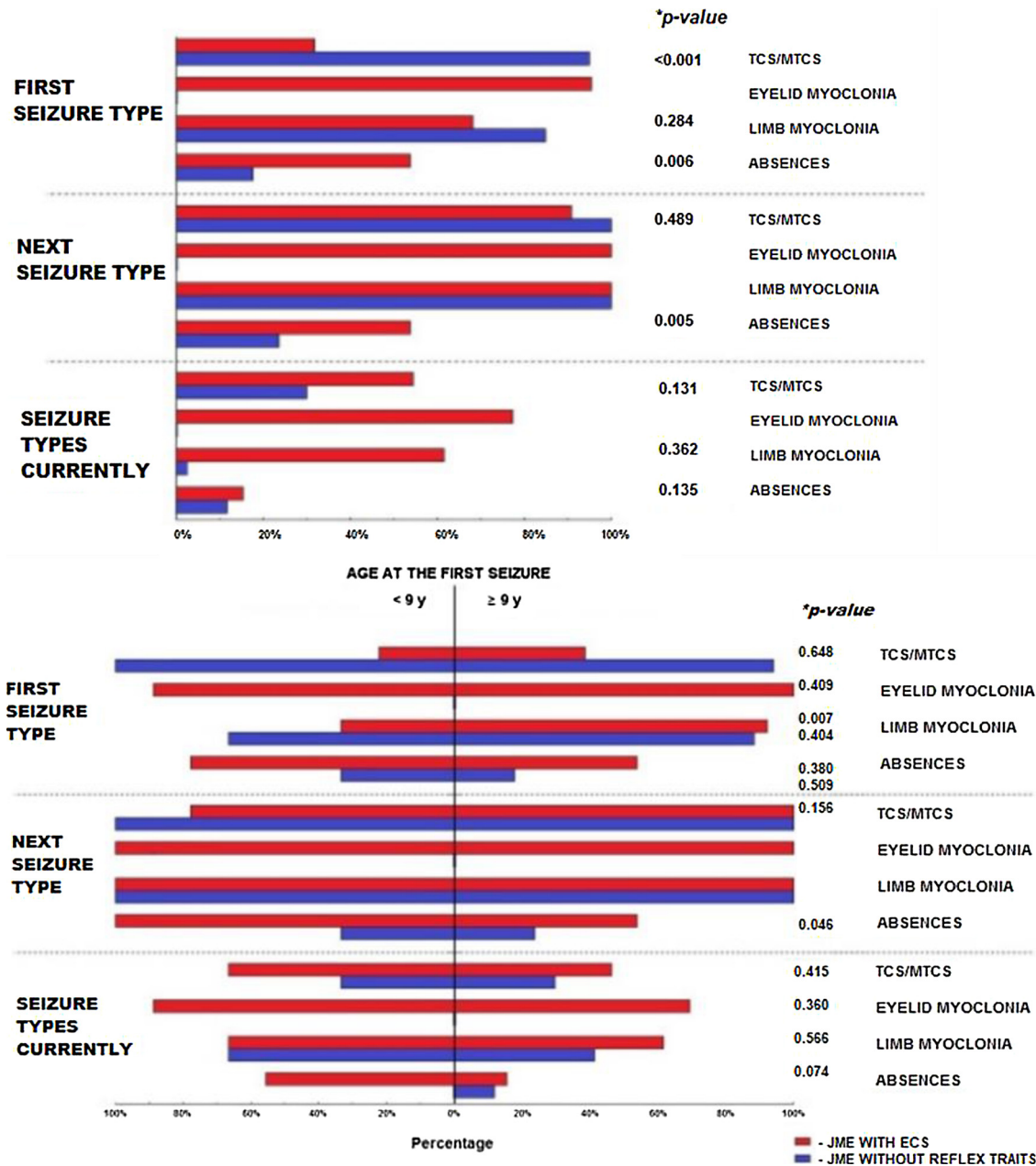


Fig. 2. Seizure patterns in patients with juvenile myoclonic epilepsy with eye closure sensitivity and in those without reflex traits since epilepsy onset, as a whole and grouped according to age at the first seizure. Legend: ECS: eye-closure sensitivity; JME: juvenile myoclonic epilepsy; TCS and/or MTCS: tonic-clonic seizures and/or myoclonic-tonic-clonic seizures. *Mann-Whitney U Test.

those reported by Caraballo et al. [21], whose series showed that 39.5% of patients with EM had familial history of epilepsy, and 25% of them had febrile seizures in childhood. Besides that, PS occurrence was significantly related to familial history of epilepsy.

4.1. Prognosis

Group 1 had a worse outcome regarding seizure control, as well as a higher frequency of limb myoclonia. While absences were controlled in almost two-thirds of patients (68.2%), EM persisted since the beginning of epilepsy in 77% of them. Moreover, 81.8% had more than 10 TCS and/or MTCS lifelong, in comparison to 50% of the patients without any reflex trait.

In this series, EM appeared as a persistent type of seizure related to earlier epilepsy onset, worse limb myoclonia control, and higher frequency of MTCS and/or TCS lifelong. However, as stated before, data

regarding EM control in JME are controversial.

EM semiological characteristics were firstly reported more than 80 years ago, by Radovici et al. [7], whereas ED and seizures triggered by eye closure began to be studied in 1955 [40–44]. The first reference to the presence of this reflex trait in JME and its prognosis was published only 34 years later, by Dalla Bernardina et al., in a series of 17 GGE patients with EM (7 with JME) [16]. According to the authors, EM would be an age-dependent electroclinical manifestation that tended to disappear between 15 and 18 years of age, even if patients were previously responding only partially to different treatments. In the series reported by these authors, even though AEDs were maintained throughout the follow-up (6–23 yrs. of age) in all JME patients, except one, the presence of EM did not modify the syndrome outcome.

On the other hand, Caraballo et al. [21] included 26 JME patients among 63 EM cases (some from Dalla Bernardina's series), and 24 of them became seizure free with VPA, ethosuximide, and

Table 2

Video-EEG data of patients with juvenile myoclonic epilepsy with eye closure sensitivity or without any reflex traits.

		JME with eye closure sensitivity (n = 22)	JME without reflex traits (n = 20)	*p
AEDs	None	2 (9.1%)	0 (0%)	0.191
	Monotherapy	14 (63.6%)	18 (90.0%)	
	Polytherapy	6 (27.3%)	2 (10.0%)	
Baseline	ED	20 (90.9%)	3 (15.0%)	< 0.001
	Myoclonia	15 (68.2%)	0 (0%)	< 0.001
Patterns of baseline ED	Spiky posterior alpha	2 (9.1%)	0 (0%)	< 0.001
	Posterior SWC	5 (22.7%)	0 (0%)	
	Diffuse SWC	6 (27.3%)	2 (10.0%)	
	Focal	0 (0%)	0 (0%)	
Sleep	ED activation	3 (14.3%)	13 (65.0%)	0.003
Hyperventilation	ED activation	12 (57.1%)	4 (20.0%)	0.025
	EM activation	9 (42.9%)	0 (0%)	< 0.001
Awakening	ED activation	6 (28.6%)	0 (0%)	< 0.001
	EM activation	10 (47.6%)	0 (0%)	< 0.001

Legend: AEDs: antiepileptic drugs; EM: eyelid myoclonia; ED: epileptiform discharges; JME: juvenile myoclonic epilepsy; SWC: spike and wave complexes.

*Fisher's Exact Test.

Bold values = $p < 0.05$.

Table 3

Specific tasks inducing eyelid myoclonia and/or epileptic discharges in patients with juvenile myoclonic epilepsy with eye closure sensitivity.

	Eyelid myoclonia		Epileptiform discharges	
	Total (%)	Activation (%)	Total (%)	Activation (%)
N	22 (100.0)		22 (100.0)	
Baseline	15 (68.2)		20 (90.9)	
Eye closure by command	15 (68.2)	9 (40.9)	11 (50.0)	7 (31.8)
Eye closure self-paced	9 (42.9)	4 (19.0)	12 (57.1)	8 (38.1)
Eye closure with blue lenses	6 (30.0)	3 (15.0)	10 (45.5)	6 (30.0)
Eye closure in darkness	0 (0)	0 (0)	0 (0)	0 (0)
Talking	10 (45.5)	8 (36.4)	7 (31.8)	4 (18.2)
Writing	4 (18.2)	4 (18.2)	4 (18.2)	1 (4.8)
Reading Portuguese aloud	2 (9.5)	2 (9.5)	4 (19.0)	2 (9.5)
Reading English aloud	3 (14.3)	2 (9.5)	5 (23.8)	2 (9.5)
Reading Portuguese silently	2 (9.1)	1 (4.8)	1 (4.5)	0 (0)
Reading English silently	1 (4.8)	1 (4.8)	1 (4.8)	1 (4.8)
Mental calculation	1 (4.8)	1 (4.8)	2 (9.5)	1 (4.8)
Eating	9 (42.9)	6 (28.6)	6 (28.6)	4 (19.0)

Legend: ED: epileptiform discharges; EM: eyelid myoclonia; PS: photosensitivity. VNPP: video-EEG neuropsychological protocol. "Activation": increase of at least twice the number of seizures or discharges compared to the baseline; *Fisher's exact test; **Mann-Whitney U test.

benzodiazepines administered as mono- or polytherapy. They found that EM in JME were usually controlled before the other seizure types, at the mean age of 12.5 yrs. (3–20 yrs.).

Two other studies reported seizure prognoses in all included JME patients with EM. Destina Yalçın et al. [19] described four female patients with EM and PS, associated with absences, myoclonic jerks causing fallings, and rare generalized tonic-clonic seizures. All patients had good school performance and total seizure control under VPA (1000–1500 mg/day). Sevgi et al. [20] included, retrospectively, 26 adult patients with ECS on EEG recordings, 6 of them with JME. Among the latter, 4 had seizure control with VPA monotherapy, and one with barbiturates. One patient lost follow-up. They concluded that the

prognosis of epilepsy in patients with ECS is excellent and seizures cease after AEDs therapy, implying that ECS is responsive to drugs.

Güvelli et al. [23] included 76 JME patients presenting ECS in EEG recordings, 12 of them with "poor prognosis" (more than one TCS per year, and/or more than 5 myoclonic or absence seizures/clusters per month, despite regular use of appropriate high doses of AEDs). Fifty out of 76 patients were taking VPA as monotherapy, and 11 as polytherapy; 10 used other AED combinations, and 5 were not treated. They concluded that ECS is a rare trait in JME, which "does not appear to be a marker of poor prognosis."

Finally, Guaranha et al. [22] and Senol et al. [24] linked EM/ECS with a worse prognosis. In their series, Guaranha et al. [22] included 65 JME patients treated with AEDs (81.6% on VPA monotherapy, 15.8% on topiramate, and 2.6% on benzodiazepines) 13 of whom had EM/ECS documented in a VNPP protocol. In this subgroup, only 4 patients (30.8%) achieved control of the three seizure types. All patients with ECS were not seizure-free. In a more recent series, Senol et al. [24] recruited 61 adult patients with EM (20 with GGEs) and classified them into three groups. Despite using appropriate and adequate AEDs (50% of them were under polytherapy), seizures, including TCS, persisted in 16 (84%) of them. Their data indicate that EM persist in adult life, may be refractory to AEDs, and require lifelong therapy.

Our results are in line with those of the latter authors [22,24]. However, our series included patients with EM documented in VNPP and followed prospectively, while all the other authors, except Guaranha et al. [21], defined EM based on standard EEGs in patients followed retrospectively [16, 19–24].

4.2. Eye closure sensitivity and photosensitivity

Of the ECS patients, 17 reported environmental PS and only 13 were photosensitive on VNPP. It is possible that the non-confirmation of photosensitivity in VNPP in these four patients may have been due to age at which the video-EEG tests were performed, since PS is a feature that becomes less frequent over the years. It is important to re-emphasize that none of the patients in Group 2 reported neither PS nor other reflex traits.

There is a considerable overlap between eye-closure sensitivity and photosensitivity, but these two reflex traits seem to be independent and develop differently over time [45]. A study regarding the visual system in EMA states that there is a "strict link" between ECS and PS, but all the 15 included cases with EMA were photosensitive, compared to 14 non-photosensitive patients with absence epilepsies and to 16 healthy controls. The EMA patients differed from both other groups by BOLD signal activation of the occipital cortex immediately upon closing the eyes, regardless of whether or not the closing of the eyes had triggered epileptiform discharges or not [46]. The authors interpreted that this finding would indicate an intrinsic functional abnormal state after eye-closure. However, it is unknown if the occipital activation relates to ECS or to PS since all patients had both traits [47]. ECS is only provoked by slow eye closure, which is generated in the supplementary motor area (SMA), and never by physiological blinking or by nociceptive reflex blinks. Therefore, the SMA has a central role in the ictogenesis and its interaction with the visual system still remains to be clarified [48].

4.3. Eye closure sensitivity in Juvenile myoclonic epilepsy and Jeavons syndrome

Regarding the prognosis in JS seizures commonly persist through a patient's lifetime, and many of them remain drug-resistant throughout life [49], similarly to what occurs with ECS in JME in our series. A recent study, including 30 patients with diagnostic criteria of JS, emphasized that after two years of follow-up, 80% of patients had drug-resistant epilepsy [50]. Compared with our data and again similar to ECS in JME, there was a diagnosis delay and a high incidence of generalized TCS over the follow-up.

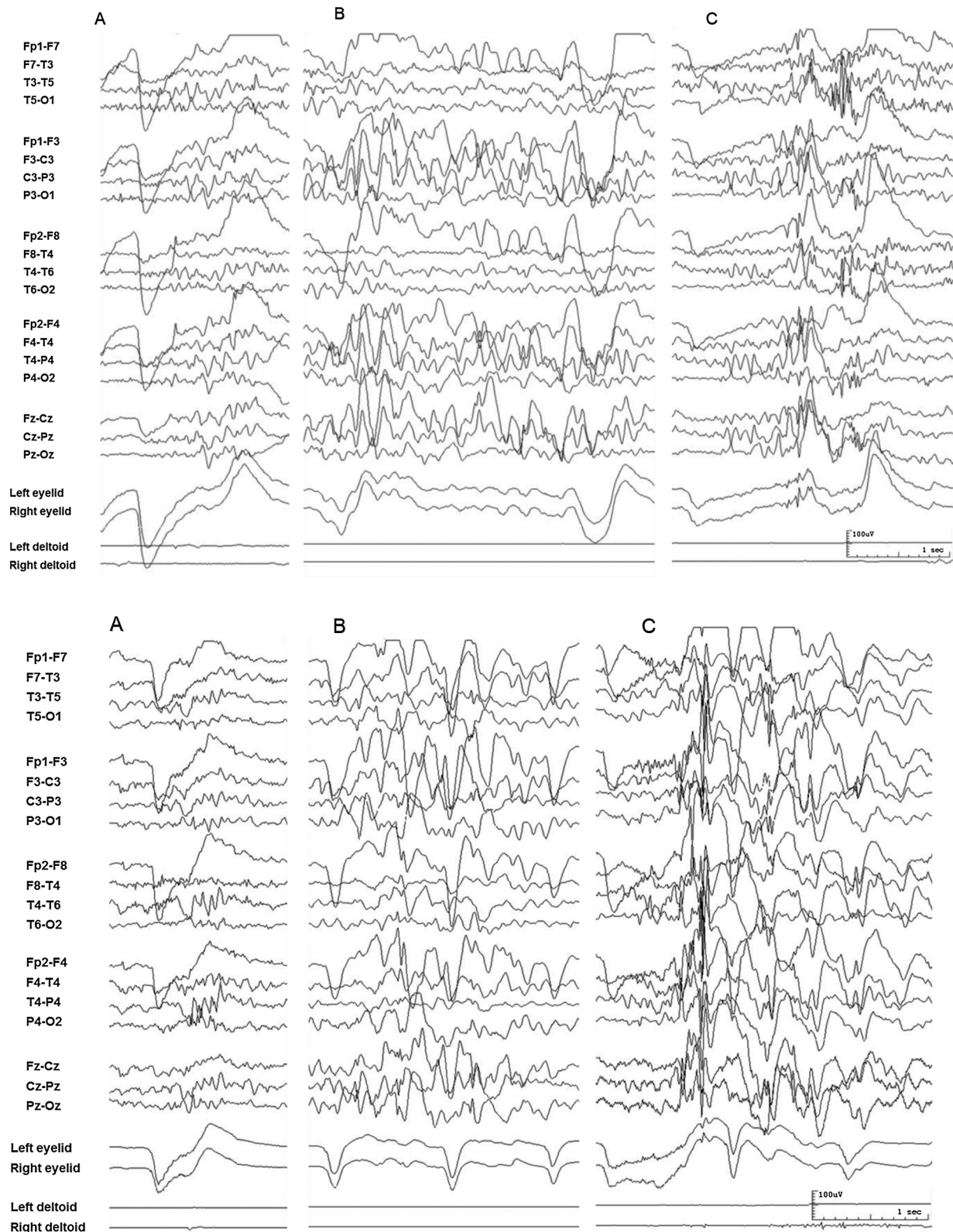


Fig. 3. Samples of eyelid myoclonia triggered by a verbal command to close the eyes. Legend: Samples of two 15-year-old females, on eye-closure under a verbal command. A. Flicker: the downward deflection phase of the eye-closure artifact has around 150 ms. of duration, and the “plateau”, around 300 ms. There is a slight tremor of the eyelids. EEG shows a short burst of “spike posterior alpha” (concomitant with the plateau). B. Flutter: the downward deflection phase of the eye-closure artifact has around 200 ms. of duration, and during the “plateau” (around 300 ms.), there is an eyelid contraction, followed by rhythmic movements of the eyelid up and down, of variable duration. EEG shows irregular diffuse spike and wave complexes. C. Jerk: the downward deflection phase of the eye-closure artifact has 700–800 ms. of duration, after which the eyes open quickly and stare. EEG shows high amplitude diffuse irregular spike and wave complexes.

Table 4

The frequency of different seizure types at epilepsy onset and currently, antiepileptic drugs intake, and psychiatric disorders in patients with juvenile myoclonic epilepsy with eye closure sensitivity or without reflex traits.

		JME with eye closure sensitivity	JME without any reflex trait	<i>p</i>
N		22	20	
Seizure types	Limb myoclonia/month (average; $\pm$ SD) at epilepsy onset	10.7 $\pm$ 10.48	13.5 $\pm$ 11.93	**0.346
	Limb myoclonia freedom currently	5 (22.7%)	11 (55.0%)	*0.055
	Limb myoclonia/month (average; $\pm$ SD) currently	6.1 $\pm$ 8.64	1.6 $\pm$ 2.89	**0.015
	Absences at epilepsy onset	14 (63.6%)	5 (25.0%)	**0.016
	Absences/month (average; $\pm$ SD) at epilepsy onset	7.9 $\pm$ 10.11	3.3 $\pm$ 9.17	**0.011
	Absences freedom currently	15 (68.2%)	18 (90.0%)	*0.135
	Absences/month (average; $\pm$ SD) currently	9.1 $\pm$ 3.81	3.0 $\pm$ 1.41	**0.083
	MTCS/month (average; $\pm$ SD) at epilepsy onset	0.9 $\pm$ 0.47	0.9 $\pm$ 1.12	**0.449
	MTCS over the last year (average; $\pm$ SD)	2.6 $\pm$ 3.97	2.7 $\pm$ 1.63	**0.673
	MTCS freedom currently	10 (45.5%)	14 (70.0%)	*0.131
	MTCS lifelong > 10	18 (81.8%)	10 (50.0%)	**0.013
	Freedom of all seizures	4 (18.2%)	9 (45.0%)	**0.023
Currently AEDs	Patients on AEDs	22 (100.0%)	20 (100.0%)	1.0
	AEDs monotherapy	15 (68.2%)	18 (90.0%)	*0.255
	Valproate use	18 (81.8%)	18 (90.0%)	**0.665
	Phenobarbital	1 (4.5%)	0 (0.0%)	
	Lamotrigine	4 (18.2%)	2 (10.0%)	**0.665
	Topiramate	4 (18.2%)	2 (10.0%)	**0.655
	Benzodiazepines	5 (22.7%)	0 (0.0%)	
	AEDs load (average; $\pm$ SD)	0.86 $\pm$ 1.023	0.70 $\pm$ 0.338	**0.263
Psychiatric disorders	Psychiatric diagnosis	8 (36.4%)	6 (30.0%)	*1.0
	Affective disorders	8 (36.4%)	4 (20.0%)	*0.738
	Psychotic disorders	0 (0.0%)	2 (10.0%)	

Legend: AEDs: antiepileptic drugs; JME: juvenile myoclonic epilepsy; MTCS: myoclonic-tonic-clonic seizures. *Fisher's exact test; **Mann-Whitney U test. Bold values = $p < 0.05$.

4.4. Eye closure sensitivity occurrence in Juvenile myoclonic epilepsy

The occurrence of EM in JME is possibly underestimated since only 18.2% of our patients reported self-perception of EM at the clinical interview and/or questionnaire. This percentage is in line with Striano et al. [11] and Covanis [12]. In contrast with Joshi and Patrick [51], who suggested that standard EEG would be sufficient for EM diagnosis, the brief duration of EM, as well as their subtle semiological manifestation, would make this condition easily overlooked or misinterpreted in routine EEG [9,52]. One pitfall for the recognition of this epileptic trait is that the room needs to be illuminated since it is inhibited in darkness environments - a condition usually presents in standard EEGs [10,20,21]. In addition, the EEG abnormalities accompanying EM are heterogeneous [52]. The correlation of EEG with EM is variable from no changes in milder, "abortive" EM [10] to generalized spike and polyspike and wave complexes [35]. In this sense, we strongly disagree with Striano and Striano's assertion [53] that "some patients may show persistence of eyelid fluttering even without concomitant EEG abnormalities, probably as a behavioral habit in subjects with a previous history of self-induced seizures." It would be risky to assume that EM is not epileptic in nature because there are no EEG changes [10,54]. An intermediary picture is the "spiky posterior alpha", that differs from the "squeak effect" reported in normal subjects. While in the latter the frequency of the alpha rhythm increases 2 Hz for 1–2 s immediately following eye closure [55] the "spiky posterior alpha" includes spike components, irregularities, and a higher amplitude sometimes spreading to the frontal regions associated with sustained eye closure [40,56].

An interesting point regarding VNPP in our patients with ECS and PS is that the propensity to activation of ED and/or seizures was increased by awakening after sleep deprivation and during eating. Probably, synkinesis of associated movements between cranial nerves and stretched spindles of oral muscles may synchronize oral-lid movements [35]. The patterns of activation could be an additional pitfall in the diagnosis of ECS reflex trait in standard EEG.

Finally, genetic confounding factors might be responsible, at least in

part, for these contradictory results regarding ECS in JME outcome. JME genetics is complex and heterogeneous and still remains not completely clarified [26]. Within the JME spectrum, some endophenotypes as PS might be defined as intermediate genetic components of diseases with complex genetics [57]. Genes related or not to ion channels have been identified, like CACNB4, GABRA1, and EFHC1, described in some families with JME, as well as SNP alleles in BRD2 [57].

Photosensitive seizures can occur in specific epilepsy syndromes, in association with other seizure types, such as JME. The photoparoxysmal response is highly heritable [58–61]. A study hypothesized that CHD2 disruption would be associated with common forms of photosensitive epilepsy or photosensitivity manifesting as a photoparoxysmal response alone [62]. The authors identified CHD2 as a photosensitive epilepsy gene. CHD2 is also the first gene to be discovered for EMA, and they suggested that CHD2 is an important contributor to both, the absence seizures with eyelid myoclonia seizure type and JS [62]. Genetic background may influence the clinical manifestations, AEDs response, and outcomes, both in the comparison of JME patients with and without reflex traits, and among those from different populations around the world.

5. Conclusions

Patients with ECS reflex trait had a lower age at epilepsy onset and a higher incidence of a familial history of epilepsy and febrile seizures.

PS was not an independent variable for prognosis, and the psychiatric comorbidity did not have prognostic implications.

In this series, ECS was related to worse outcome regarding control of all seizure types, the persistence of EM in around two-thirds of patients, and a higher frequency of limb myoclonia currently, as well as of the total number of TCS and/or MTCS.

Limitations of the study

Our study was conducted in a tertiary specialized epilepsy center,

whose patients may not represent the general population.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Conflict of interest statement

None of the authors has any conflict of interest to disclose.

Acknowledgments

This study belongs to the EpExMo cooperation, an open international research initiative for the study of epilepsies with external modulation of ictogenesis. The participating centers in this study were: Universidade Federal de São Paulo-Escola Paulista de Medicina, São Paulo, Brazil (C.G.P. Uchida, M.S.B. Guaranha, K.C. Carvalho, L.M.F.F. Guilhoto, G.M. De Araújo Filho, E.M.T. Yacubian) and Danish Epilepsy Centre, Dianalund, Denmark (P. Wolf). The participation of P. Wolf was made possible by the Ciência sem Fronteiras Program of CAPES, Brazil.

References

- Jallon P, Latour P. Epidemiology of idiopathic generalized epilepsies. *Epilepsia* 2005;46(Suppl. 1):10–4.
- Genton P, Thomas P, Kasteleijn-Nolst Trenité DG, Medina MT, Salas-Puig J. Clinical aspects of juvenile myoclonic epilepsy. *Epilepsy Behav* 2013;28(Suppl. 1):8–14.
- Camfield CS, Striano P, Camfield PR. Epidemiology of juvenile myoclonic epilepsy. *Epilepsy Behav* 2013;28(Suppl. 1):15–7.
- Fisher RS, Cross JH, French JA, Higurashi N, Hirsch E, Jansen FE, et al. Operational classification of seizure types by the international league against epilepsy: position paper of the ILAE commission for classification and terminology. *Epilepsia* 2017;58(4):522–30.
- Fisher RS, Cross JH, D'Souza C, French J, Haut SR, Higurashi N, et al. Instruction manual for the ILAE 2017 operational classification of seizure types. *Epilepsia* 2017;58(4):531–2.
- Duncan JS, Panayiotopoulos CP. The differentiation of 'eye closure' from 'eyes-closed' EEG abnormalities and their relation to photo- and fixation-off sensitivity. In: Duncan JS, Panayiotopoulos CP, editors. *Eyelid myoclonia with absences*. London: John Libbey; 1996. p. 77–87.
- Radovici MMA, Misirlivi VL, Gluckman M. Epilepsy reflex provoquée par excitations optiques des rayons solaires. *Rev Neurol (Paris)* 1932;1:1305–7.
- Jeavons PM. Nosological problems of myoclonic epilepsies in childhood and adolescence. *Dev Med Child Neurol* 1977;19:3–8.
- Appleton RE, Panayiotopoulos CP, Acomb BA, Beirne M. Eyelid myoclonia with typical absences: an epilepsy syndrome. *J Neurol Neurosurg Psychiatry* 1993;56(12):1312–6.
- Giannakodimos S, Panayiotopoulos CP. Eyelid myoclonia with absences in adults: a clinical and video-EEG study. *Epilepsia* 1996;37(1):36–44.
- Striano S, Striano P, Nocerino C, Boccella P, Bilo L, Meo R, et al. Eyelid myoclonia with absences: an overlooked epileptic syndrome? *Neurophysiol Clin* 2002;32(5):287–96.
- Covanis A. Photosensitivity in idiopathic generalized epilepsies. *Epilepsia* 2005;46(Suppl. 9):67–72.
- Engel Jr J, International League Against Epilepsy (ILAE). A proposed diagnostic scheme for people with epileptic seizures and with epilepsy: report of the ILAE Task force on classification and terminology. *Epilepsia* 2001;42(6):796–803.
- Ferrie CD, Agathonikou A, Parker A, Robinson RO, Panayiotopoulos CP, Panayiotopoulos CP. The spectrum of childhood epilepsies with eyelid myoclonia. In: Duncan JS, editor. *Eyelid myoclonia with absences*. London: John Libbey; 1996. p. 39–48.
- Panayiotopoulos CP, Agathonikou A, Koutroumanidis M, Giannakodimos S, Rowlinson S, Carr CP. Eyelid myoclonia with absences: the symptoms. In: Duncan JS, Panayiotopoulos CP, editors. *Eyelid myoclonia with absences*. London: John Libbey; 1996. p. 17–26.
- Dalla Bernardina B, Sgro V, Fontana E, Cellino MR, Moser C, Zullini E, et al. Eyelid myoclonia with absences. In: Beaumanoir A, Gastaut H, Naquet R, editors. *Reflex seizures and reflex epilepsies*. Geneva: Medicine et Hygiene; 1989. p. 193–200.
- Baykan-Kurt B, Gökyigit A, Parman Y, Kinay D, Gürses C. Eye closure related spike and wave discharges: clinical and syndromic associations. *Clin Electroencephalogr* 1999;30(3):106–10.
- Panayiotopoulos CP. Syndromes of idiopathic generalized epilepsies not recognized by the international league against epilepsy. *Epilepsia* 2005;9:57–66.
- Destina Yalçın A, Forta H, Kiliç E. Overlap cases of eyelid myoclonia with absences and juvenile myoclonic epilepsy. *Seizure* 2006;15(6):359–65.
- Sevgi EB, Saygi S, Ciger A. Eye closure sensitivity and epileptic syndromes: a retrospective study of 26 adult cases. *Seizure* 2007;16(1):17–21.
- Caraballo RH, Fontana E, Darra F, Chacon S, Ross N, Fiorini E, et al. A study of 63 cases with eyelid myoclonia with or without absences: type of seizure or an epileptic syndrome? *Seizure* 2009;18(6):440–5.
- Guaranha MS, Filho GM, Lin K, Guilhoto LM, Caboclo LO, Yacubian EM. Prognosis of juvenile myoclonic epilepsy is related to endophenotypes. *Seizure* 2011;20:42–8.
- Güveli BT, Baykan B, Dörtcan N, Bebek N, Gürses C, Gökyigit A. Eye closure sensitivity in juvenile myoclonic epilepsy and its effect on prognosis. *Seizure* 2013;22:867–71.
- Nar Senol P, Tezer FI, Saygi S. Eyelid myoclonia seizures in adults: an alternate look at the syndrome paradox. *Epilepsy Behav* 2015;45:265–70.
- Kasteleijn-Nolst Trenité DG, Schmitz B, Janz D, Delgado-Escueta AV, Thomas P, Hirsch E, et al. Consensus on diagnosis and management of JME: from founder's observations to current trends. *Epilepsy Behav* 2013;28(Suppl. 1):87–90.
- Wolf P, Yacubian EM, Avanzini G, Sander T, Schmitz B, Wandschneider B, et al. Juvenile myoclonic epilepsy: a system disorder of the brain. *Epilepsy Res* 2015;114:2–12.
- Sheehan DV, Lecrubier Y, Sheehan KH, Amorim P, Janavs J, Weiller E, et al. The Mini-International Neuropsychiatric Interview (M.I.N.I.): the development and validation of a structured diagnostic psychiatric interview for DSM-IV and ICD-10. *J Clin Psychiatry* 1998;59(Suppl. 20):22–33.
- Biaggio AMB, Natalício L, Spielberger C. Desenvolvimento da forma experimental em português do Inventário de Ansiedade Traço-Estado de Spielberger. *Arq Bras Psicol Aplicada* 1977;29(3):31–44.
- Matsuoka H, Takahashi T, Sasaki M, Matsumoto K, Yoshida S, Numachi Y, et al. Neuropsychological EEG activation in patients with epilepsy. *Brain* 2000;123:318–30.
- Guaranha MSB, Silva Sousa P, De Araújo-Filho GM, Lin K, Guilhoto LM, Caboclo LO, et al. Provocative and inhibitory effects of a video-EEG neuropsychologic protocol in juvenile myoclonic epilepsy. *Epilepsia* 2009;50:2446–55.
- Siivola J, Saarinen A. Piezoelectric film transducer for recording of oculography in electro-encephalogram-polygraphy. *Med Biol Eng Comput* 2002;40(4):469–70.
- Kasteleijn-Nolst Trenité D, Rubboli G, Hirsch E, Martins da Silva A, Seri S, Wilkins A, et al. Methodology of photic stimulation revisited: updated European algorithm for visual stimulation in the EEG laboratory. *Epilepsia* 2012;53(1):16–24.
- Capovilla G, Gambardella A, Rubboli G, Beccaria F, Montagnini A, Aguglia U, et al. Suppressive efficacy by a commercially available blue lens on PPR in 610 photosensitive epilepsy patients. *Epilepsia* 2006;47(3):529–33.
- Covanis A. Jeavons syndrome. In: Panayiotopoulos CP, editor. *Atlas of epilepsies*. London: Springer-Verlag; 2010. p. 1080–91.
- Da Conceição PO, Guaranha MSB, Uchida CGP, Carvalho K, Guilhoto LM, Carrete Jr H, et al. Blinking and eyelid myoclonia: characteristics and correlations of eyelid movements. *Seizure* 2015;24:12–6.
- Deckers CL, Hekster YA, Keyser A, Meinardi H, Renier WO. Reappraisal of polytherapy in epilepsy. A critical review of drug load and adverse effects. *Epilepsia* 1997;38(5):570–5.
- World Health Organization. Lists of new ATC/DDDs and alterations. 2016 [Accessed 1 November 2016]. https://www.whocc.no/ddd/lists_of_new_atc_ddds_and_alterations/.
- Janz D. Epilepsie mit impulsiv-petit mal. In: Janz D, editor. *Die epilepsien: spezielle pathologie und therapie*. Stuttgart: Georg Thieme Verlag; 1969. p. 135–63.
- Waltz S. The EEG in juvenile myoclonic epilepsy. In: Schmitz B, Sander T, editors. *Juvenile myoclonic epilepsy: the Janz syndrome*. Petersfield: Wrightson Biomedical Publishing Ltd; 2000. p. 41–55.
- Benassi P, Corsino GM. Rapporti tra reazioni d'arresto e manifestazione epilettiche parossistiche. *Riv Sper Freniatr Med Leg Alien Ment* 1955;79(2):299–324. 30.
- Gastaut H, Tassinari CA. Triggering mechanisms in epilepsy. The electroclinical point of view. *Epilepsia* 1966;7(2):85–138.
- Green JB. Self-induced seizures. *Arch Neurol* 1966;15(6):579–86.
- Green JB. Seizures on closing the eyes. *Electroencephalographic studies*. *Neurology* 1968;18(4):391–6.
- Giovanardi Rossi P, Frank L, Pazzaglia P. Crisi epilettiche alla chiusura degli occhi. *G Psychiatr Neuropatol* 1969;97(3):469–76.
- Fabian A, Wolf P. Epileptic discharge after eye closure: relation to photosensitivity. Wolf P, Dam M, Janz D, Dreifuss FE, editors. *Advances in epileptology*, vol. 16. New York: Raven Press; 1987. p. 259–64.
- Vaudano AE, Ruggieri A, Tondelli M, Avanzini P, Benuzzi F, Gessaroli G, et al. The visual system in eyelid myoclonia with absences. *Ann Neurol* 2014;76:412–27.
- Wolf P. Reply to "comment to 'reflex epileptic mechanisms in humans: lessons about natural ictogenesis' by Peter Wolf" by Stefano Meletti and Anna Elisabetta Vaudano. *Epilepsy Behav* 2015;52:277–8.
- Wolf P. Reflex epileptic mechanisms in humans: lessons about natural ictogenesis. *Epilepsy Behav* 2017;71:118–23.
- Striano P, Sofia V, Capovilla G, Rubboli G, Di Bonaventura C, Coppola A, et al. A pilot study of levetiracetam in eyelid myoclonia with absences (Jeavons syndrome). *Epilepsia* 2008;49(3):425–30.
- Smith KM, Youssef PE, Wirrel EC, Nickels KC, Payne ET, Britton JW, et al. Jeavons syndrome: clinical features and response to treatment. *J Pediatr Neurol* 2018. <https://doi.org/10.1016/j.pediatrneurol.2018.06.001>.
- Joshi CN, Patrick J. Eyelid myoclonia with absences: routine EEG is sufficient to make a diagnosis. *Seizure* 2007;16:254–60.
- Wallace SJ. Eyelid myoclonia with absences in children: the Cardiff experience. In: Duncan JS, Panayiotopoulos CP, editors. *Eyelid myoclonia with absences*. London: John Libbey; 1996. p. 27–31.
- Striano S, Striano P. Eyelid myoclonia with absences (Jeavons syndrome): still an overlooked epilepsy syndrome. *J Epileptol* 2015;23:125–7.
- Binnie CD. Differential diagnosis of eyelid myoclonia with absences and self-

- induction by eye closure. In: Duncan JS, Panayiotopoulos CP, editors. Eyelid myoclonia with absences. London: John Libbey; 1996. p. 89–92.
- [55] Bekkering DH, Kamp A, Storm van Leeuwen W. The EEG-spectrograph. *Electroenceph Clin Neurophysiol* 1958;10:555–9.
- [56] Viravan S, Go C, Ochi A, Akiyama T, Carter Snead 3rd O, Otsubo H. Jeavons syndrome existing as occipital cortex initiating generalized epilepsy. *Epilepsia* 2011;52(7):1273–9.
- [57] Baykan B, Wolf P. Juvenile myoclonic epilepsy as a spectrum disorder: a focused review. *Seizure* 2017;49:36–41.
- [58] Sadleir LG, Vears D, Regan B, Redshaw N, Bleasel A, Scheffer IE. Family studies of individuals with eyelid myoclonia with absences. *Epilepsia* 2012;53(12):2141–8.
- [59] Waltz S, Stephani U. Inheritance of photosensitivity. *Neuropediatrics* 2000;31:82–5.
- [60] Tauer U, Lorenz S, Lenzen KP, Heils A, Muhle H, Gresch M, et al. Genetic dissection of photosensitivity and its relation to idiopathic generalized epilepsy. *Ann Neurol* 2005;57(6):866–73.
- [61] Taylor I, Berkovic SF, Scheffer IE. Genetics of epilepsy syndromes in families with photosensitivity. *Neurology* 2013;80(14):1322–9.
- [62] Galizia EC, Myers CT, Leu C, de Kovel CG, Afrikanova T, Cordero-Maldonado ML, et al. CHD2 variants are a risk factor for photosensitivity in epilepsy. *Brain* 2015;138:1198–207.